



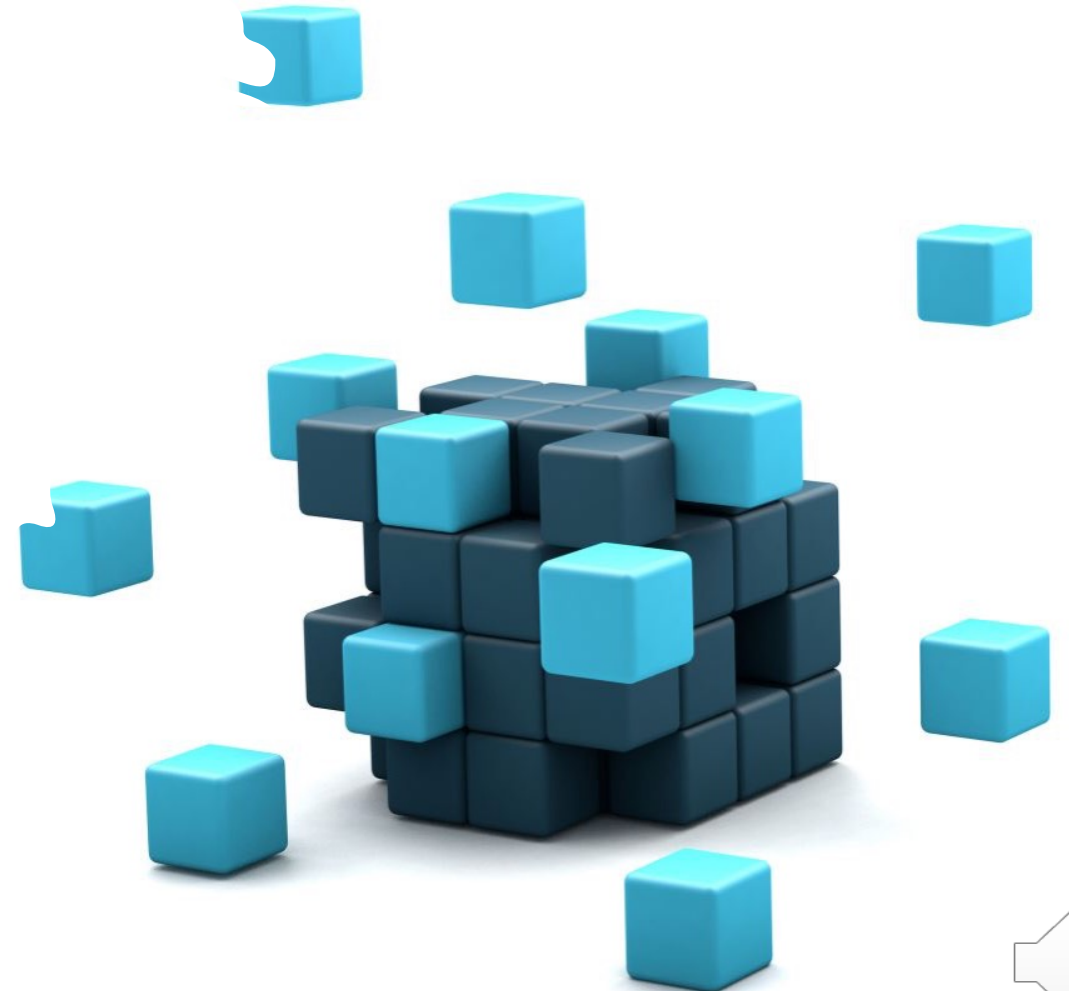
EGH404

Data Analysis: Data Wrangling, Statistics & Visualisation

Faculty of Engineering

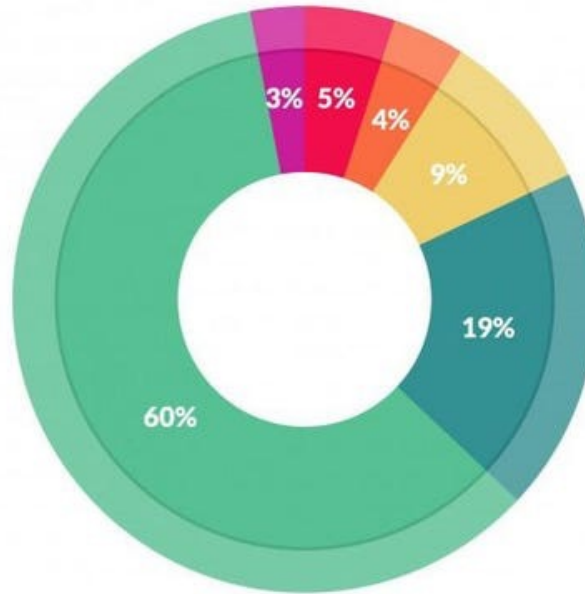


Data provides the
building blocks of
empirical
(i.e. evidence-based)
research



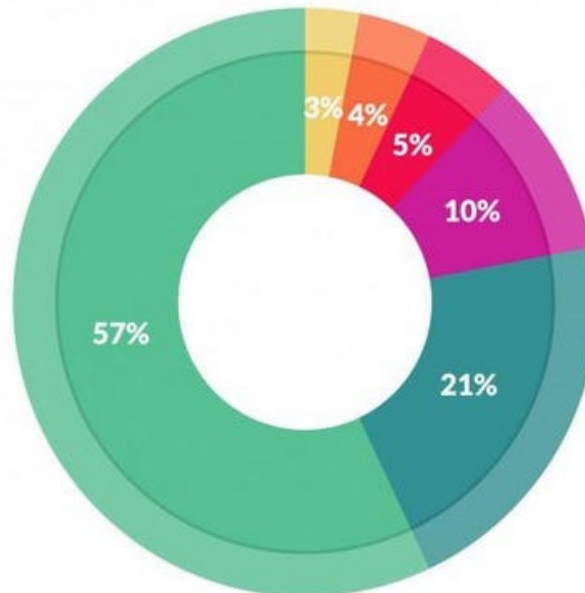
Data Wrangling

- In a report published by Forbes **data preparation accounts for about 80% of a data scientists role, with 60% of that time spent with the cleaning and organizing of that data for analysis**



What data scientists spend the most time doing

- Building training sets: 3%
- Cleaning and organizing data: 60%
- Collecting data sets: 19%
- Mining data for patterns: 9%
- Refining algorithms: 4%
- Other: 5%



What's the least enjoyable part of data science?

- Building training sets: 10%
- Cleaning and organizing data: 57%
- Collecting data sets: 21%
- Mining data for patterns: 3%
- Refining algorithms: 4%
- Other: 5%





The goal of data wrangling is to convert messy and complex data into high-quality, usable formats.

- Handling Missing Values
- Removing Duplicates
- Integrating / Merging Data
- Filtering Data
- Grouping Data
- Reshaping Data
- Transforming Data



Problematic Data (e.g. Outliers, Missing Data)

Assess on case-by-case basis

Consider how to best handle problematic data given the context of your problem

- Does the data make sense?
- Look at research articles in your field – what have they done? What is the rationale?
- Does the data follow the appropriate rules for its field? If not, is that because of a data quality issue?

Handling problematic data:

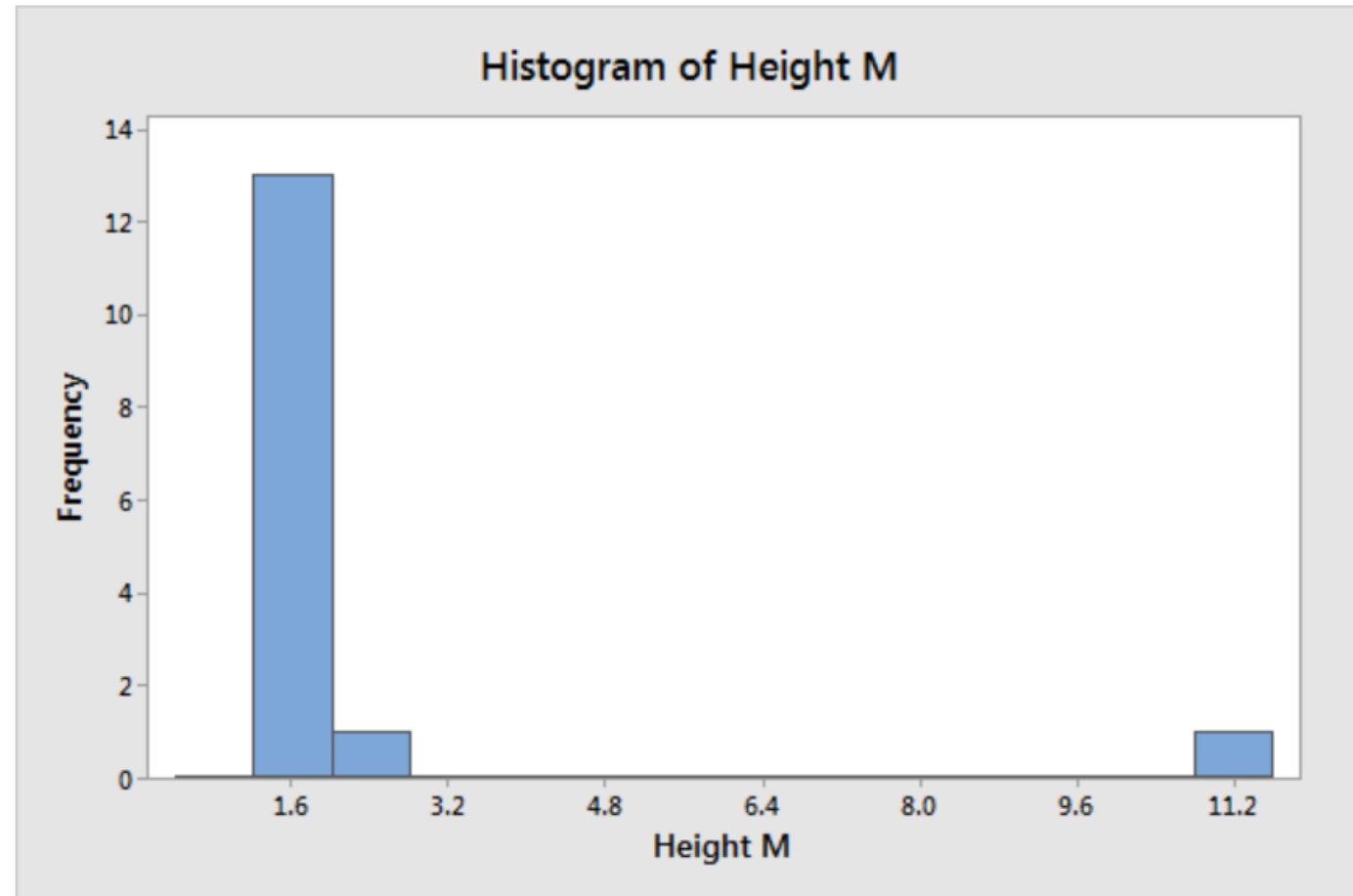
Identify most suitable option(s):

- Correct errors (e.g. “NaN”)
- Remove Duplicates and/or irrelevant observations
- Removing data (e.g. if $< 5\%$, acceptable to ignore/remove problematic data)
- Impute? (mean, median, mode)
- Set default values?
- Use predictive modeling (e.g. regression models) to replace problematic data



Identify Outliers: Plot Histograms

Histograms also emphasize the existence of outliers. Look for isolated bars, as shown below. Our outlier is the bar far to the right. The graph crams the legitimate data points on the far left.



Identify Outliers: Plot Boxplots

Use boxplot visualisations to help identify outliers (unrealistic values/anomalies)

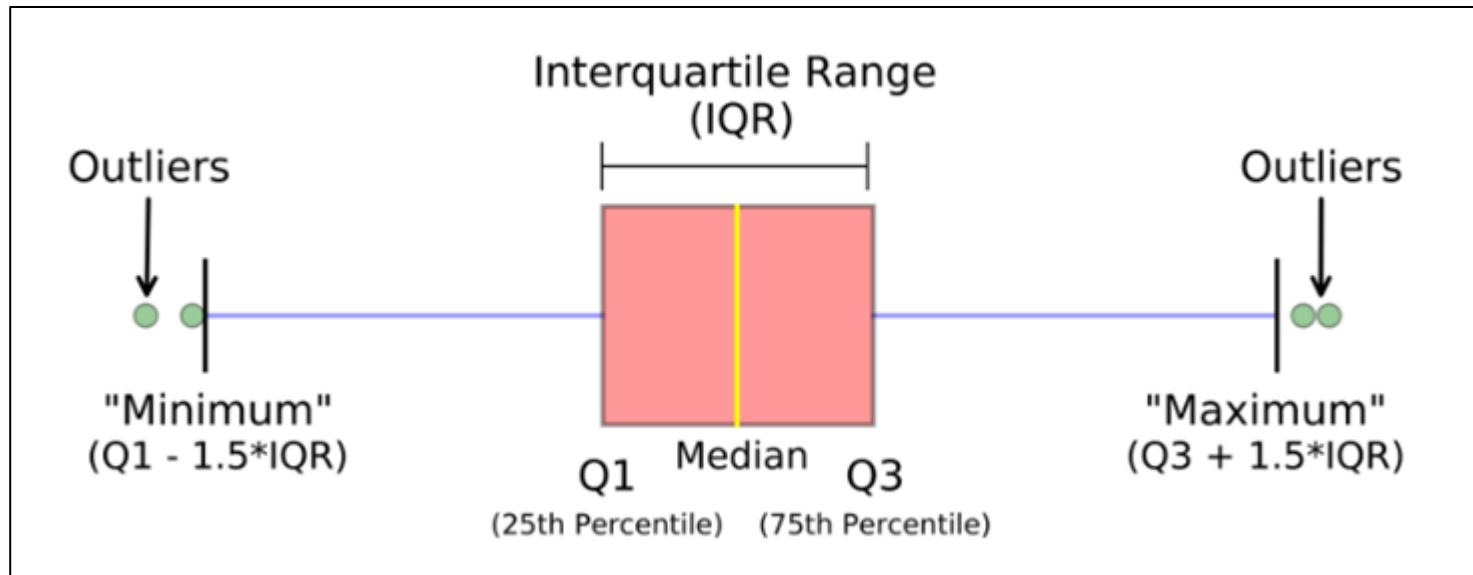
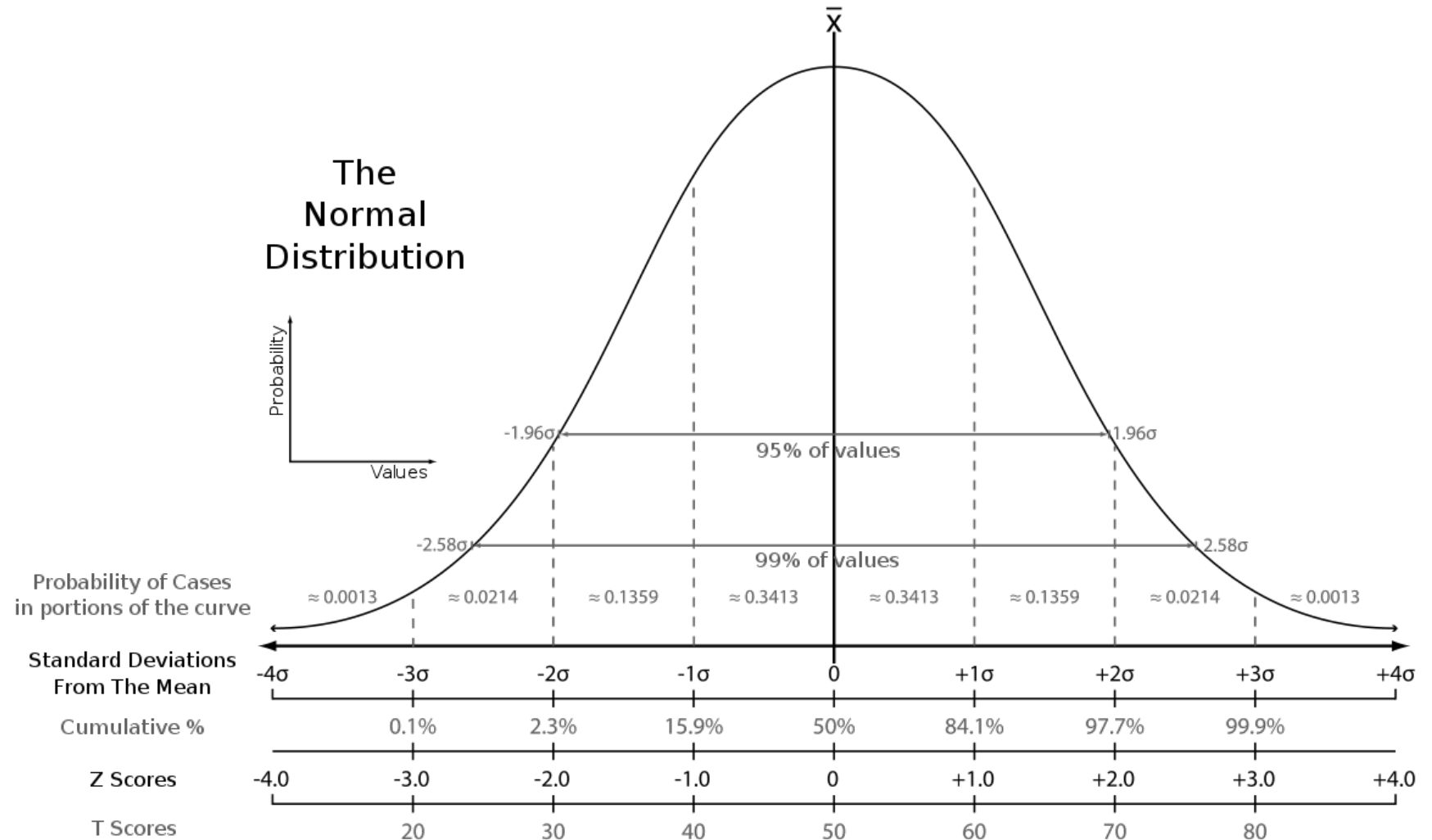


Image source: <https://www.analyticsvidhya.com/blog/2022/10/outliers-detection-using-iqr-z-score-lof-and-dbscan/>

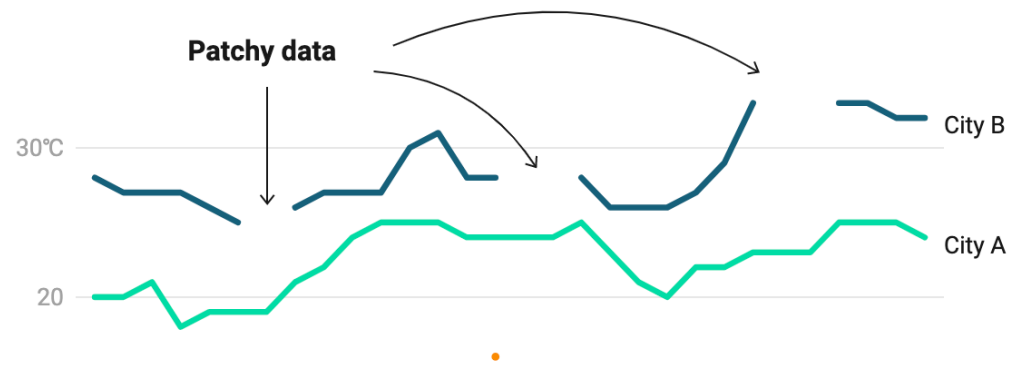
- One way to deal with outliers is to use the interquartile range:
 - Find the first quartile, Q1.
 - Find the third quartile, Q3.
 - Calculate the IQR. $IQR = Q3 - Q1$.
 - Define the normal data range with lower limit as $Q1 - 1.5 * IQR$ and upper limit as $Q3 + 1.5 * IQR$.
 - Any data point outside this range is considered as outlier and should be removed for further analysis.



Identify Outliers: standard deviations, z-scores



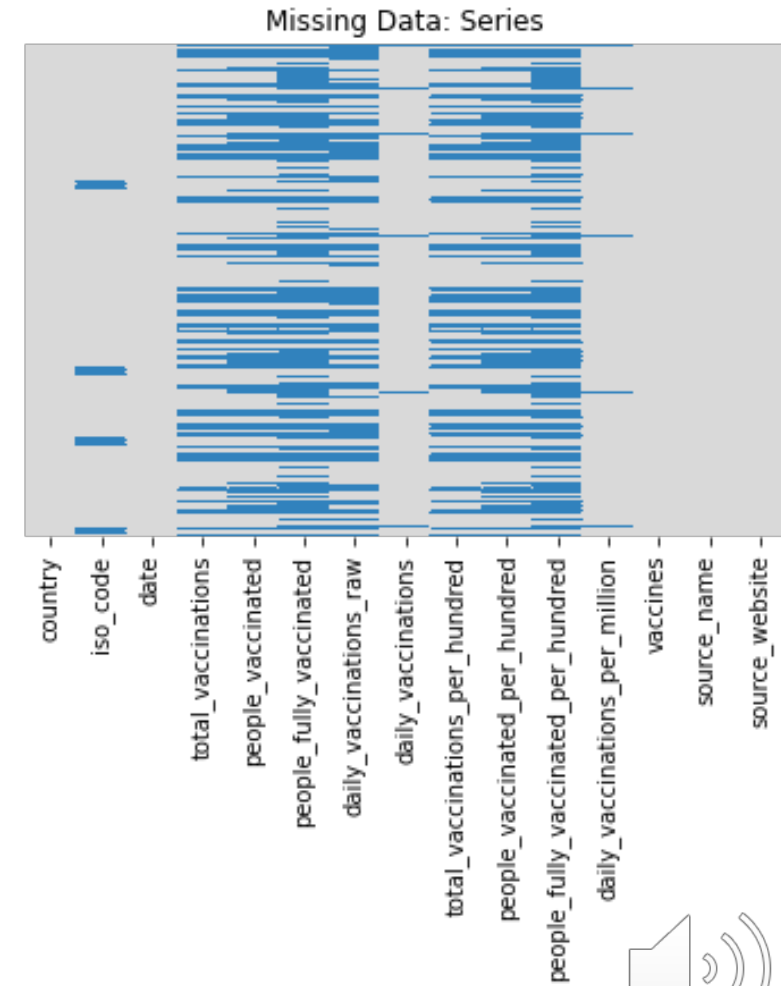
Identify Missing Data



```
rows = series.shape[0]
null_total = series.isnull().sum()
missing_percent = (null_total/rows)*100
missing_percent
```

With this code you can find the percentage of missing values

country	0.000000
iso_code	6.669517
date	0.000000
total_vaccinations	34.715690
people_vaccinated	44.527576
people_fully_vaccinated	62.676357
daily_vaccinations_raw	44.933732
daily_vaccinations	3.420265
total_vaccinations_per_hundred	34.715690
people_vaccinated_per_hundred	44.527576
people_fully_vaccinated_per_hundred	62.676357
daily_vaccinations_per_million	3.420265
vaccines	0.000000
source_name	0.000000
source_website	0.000000
dtype:	float64

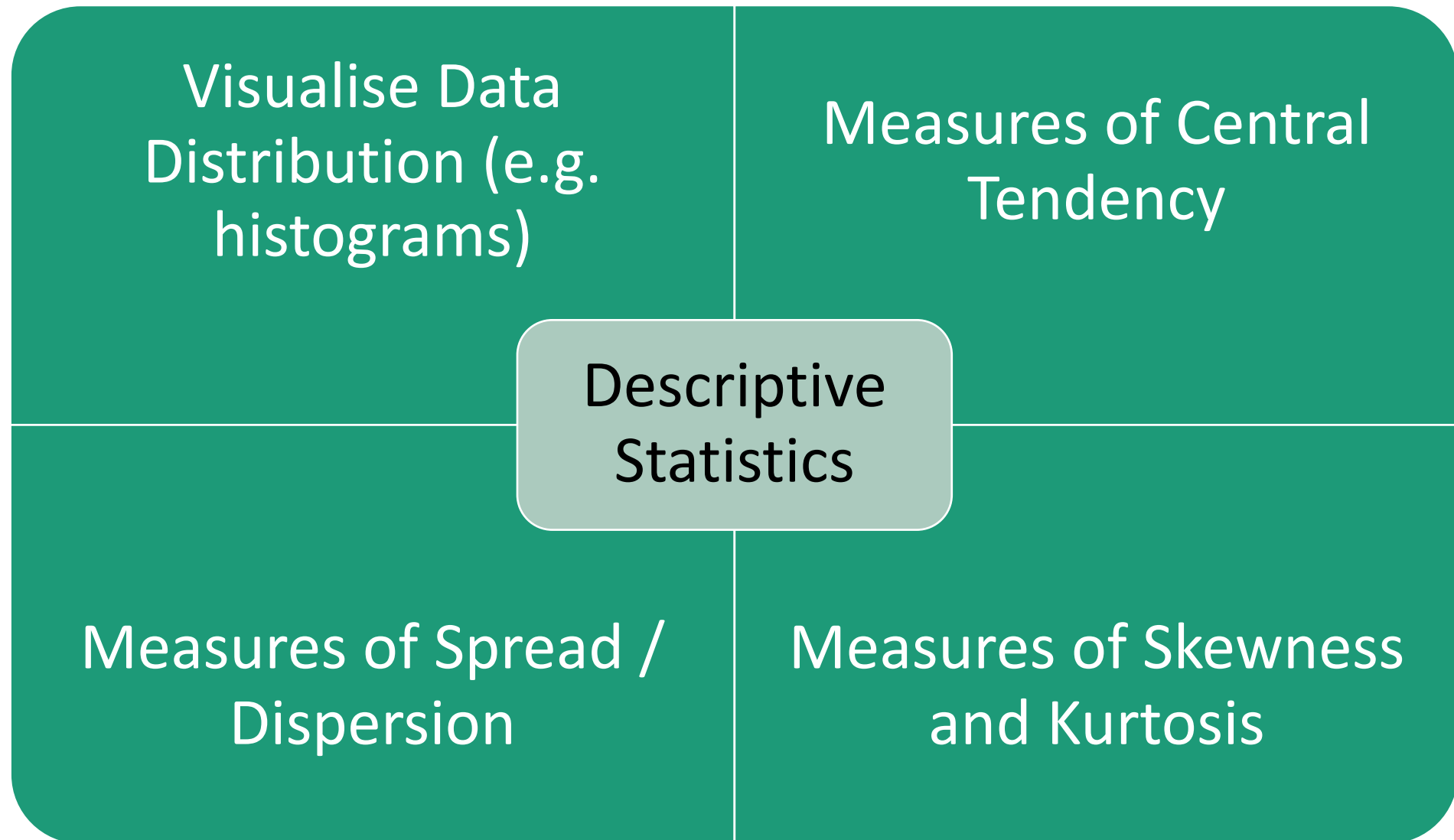


Summary Statistics

(aka Descriptive Statistics)

Please watch module materials on Summary Statistics

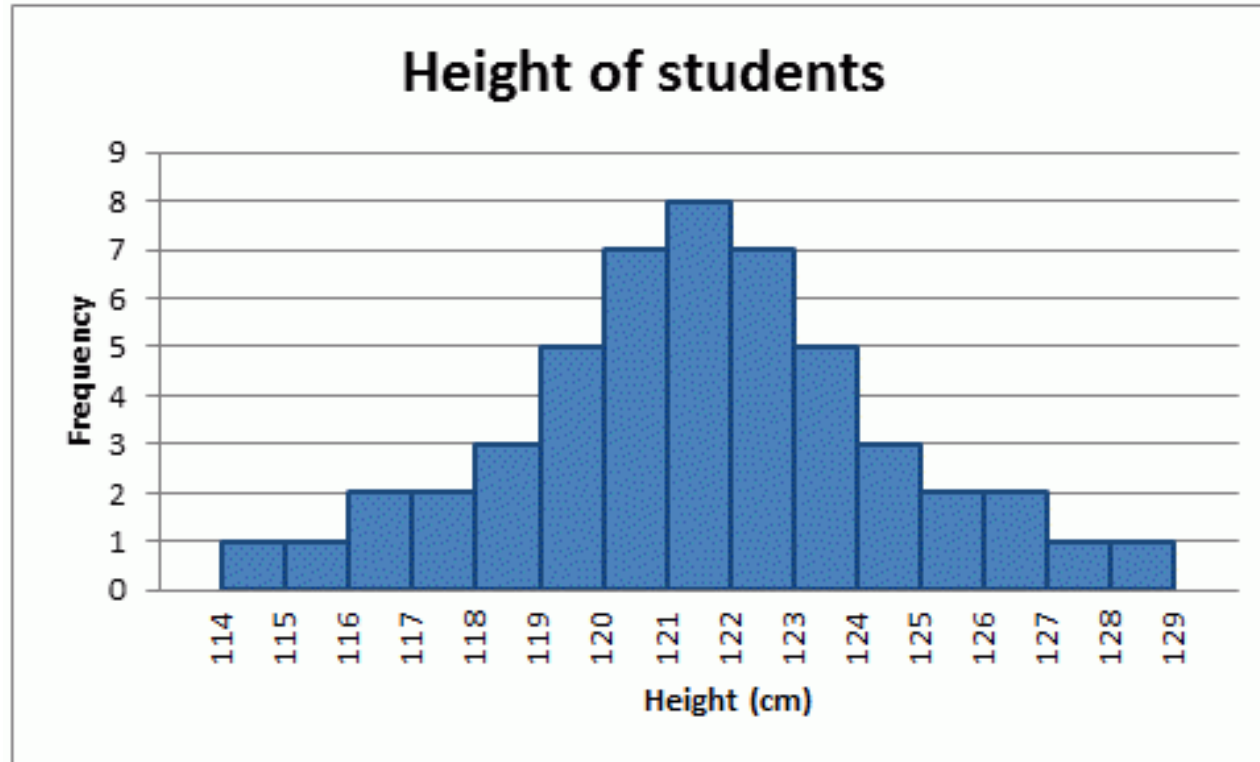




Visualise Data Distribution: Histograms

Histogram graph showing the frequency of student heights.

- 114cm - 1
- 115cm - 1
- 116cm - 2
- 117cm - 2
- 118cm - 3
- 119cm - 5
- 120cm - 7
- 121cm - 8
- 122cm - 8
- 123cm - 7
- 124cm - 5
- 125cm - 3
- 126cm - 2
- 127cm - 2
- 128cm - 1
- 129cm - 1

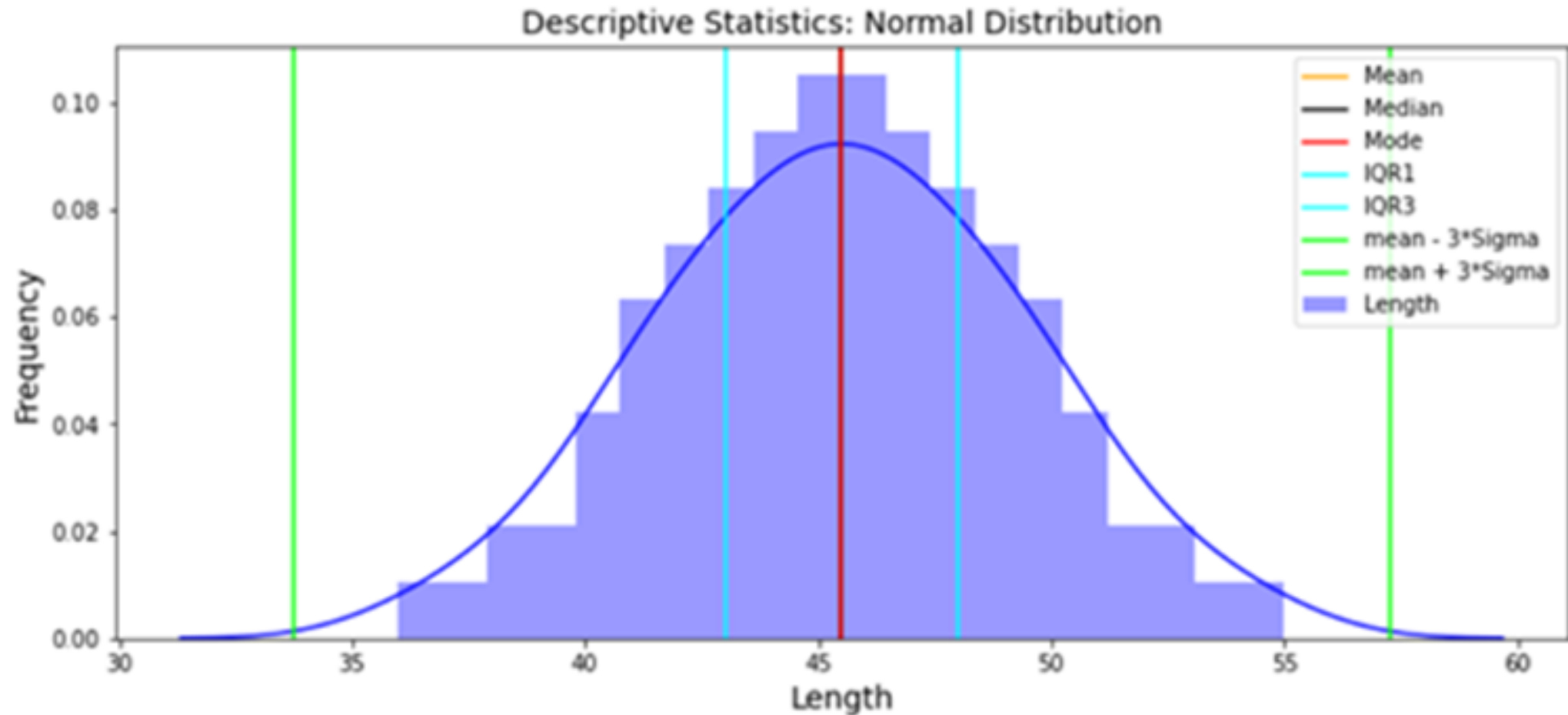


A histogram helps describe the distribution (or pattern) of the data within a dataset.

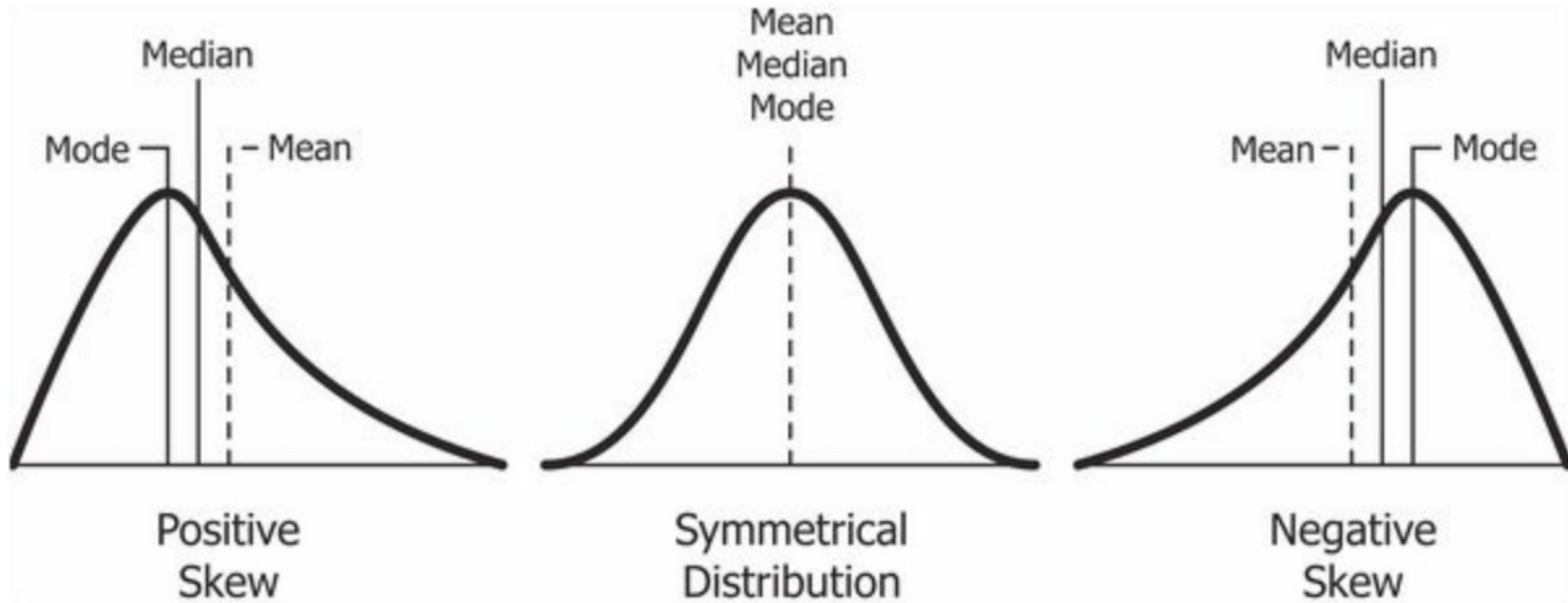


Central Tendency, Spread & Dispersion

Histograms help describe the distribution (or pattern) of the data within a dataset.



Skewness



So what?

Imagine you're creating a new kind of material to use in buildings. You're checking how strong it is by doing strength tests

E.g. Strength data (in units) collected from 30 tests = $23 + 25 + 26 + 28 + 24 + 25 + 26 + 27 + 29 + 25 + 26 + 28 + 24 + 26 + 25 + 800 + 25 + 28 + 26 + 27 + 26 + 25 + 26 + 28 + 27 + 25 + 26 + 24 + 25 + 26 = 1551$

Average = the sum (1551) divided by the total count of numbers (30)) = $1551 / 30 \approx 51.17$

Since the data set has 30 values, the median is the middle value when the data is arranged in order. In this case, the median is the 15th value, which is 26.

Skewed results make the material appear stronger than it truly is. As engineers be careful with extreme results – as they will directly impact on the safety & reliability of building design.



Recap: Always plot your data to understand your data

- Consider different ways to inspect the distribution of your data
- Plot your data using box plots, histograms, density plots, violin plots, etc.. For examples plots go to: <https://www.data-to-viz.com/story/OneNumOneCatSeveralObs.html>
- What does the distribution of your data tell you?

